# Duc A. Hoang

*Curriculum Vitae*

This CV was updated on July 6, 2024.

## Personal Information

| | |
|---|---|
| Full name (Vietnamese) | Hoàng Anh Đức. |
| Name (in publications) | Duc A. Hoang. |
| Name (Japanese Katakana) | ホアンアンドゥック. |
| Nationality | Vietnamese. |
| Gender | Male. |
| Languages | Vietnamese (Native) and English (Professional working proficiency). |

## Current Position

As of February 01, 2023, I am a **Lecturer** at Faculty of Mathematics, Mechanics, and Informatics, VNU University of Science (Hanoi, Vietnam).

## Contact Information


| | |
|---|---|
| Email | hoanganhduc@hus.edu.vn |
| Email | anhduc.hoang1990@gmail.com |
| Institution | Faculty of Mathematics, Mechanics, and Informatics, VNU University of Science, 334 Nguyen Trai, Thanh Xuan, Hanoi, 100000 Vietnam. |
| Personal Webpage | https://hoanganhduc.github.io/ |
| ORCID | 0000-0002-8635-8462 |


## Research Interests

- Graph Algorithms.
- Combinatorial Reconfiguration.

## Education

Apr. 2015 - Jun. 2018 **PhD Degree in Information Science**

- **Institution:** Japan Advanced Institute of Science and Technology (Ishikawa, Japan).
- **Supervisor:** Ryuhei UEHARA.
- **Thesis Title:** Independent Set Reconfiguration and Related Problems for Some Restricted Graphs.

Apr. 2013 - Mar. 2015 **Master Degree in Information Science**

- **Institution:** Japan Advanced Institute of Science and Technology (Ishikawa, Japan).
- **Supervisor:** Ryuhei UEHARA.
- **Thesis Title:** The Independent Set Reconfiguration Problem on Some Restricted Graphs.

Sep. 2008 - Mar. 2013 **Bachelor Degree in Mathematics**

- **Institution:** VNU University of Science (Hanoi, Vietnam).
- **Thesis Advisor:** Thi Ha Duong PHAN.
- **Thesis Title:** The Matrix-Tree Theorem and Some Related Problems.
- Advanced Undergraduate Program in Mathematics.

## Employment

Feb. 01, 2023 - present **Lecturer**
Department of Informatics, Faculty of Mathematics, Mechanics, and Informatics, VNU University of Science (Hanoi, Vietnam).

Jun. 16, 2021 - Jan. 31, 2023 **Postdoctoral Researcher**
Group of Computer Algorithms, Graduate School of Informatics, Kyoto University (Kyoto, Japan).
Supervisor: Shin-ichi MINATO.
B01 Group, AFSA Project (supported by KAKENHI Grant Number 20H05964).

Apr. 01, 2021 - Jun. 15, 2021 **Research Assistant**
Kyutech Algorithms Group, School of Computer Science and Systems Engineering, Kyushu Institute of Technology (Fukuoka, Japan).
Supervisor: Toshiki SAITOH.

Apr. 01, 2019 - Mar. 31, 2021 **Postdoctoral Researcher**
Kyutech Algorithms Group, School of Computer Science and Systems Engineering, Kyushu Institute of Technology (Fukuoka, Japan).
Supervisor: Toshiki SAITOH.

Sep. 05, 2018 - Dec. 31, 2018 **Lecturer**
Department of Informatics, Faculty of Mathematics, Mechanics, and Informatics, VNU University of Science (Hanoi, Vietnam).

## Research Grants

Jul. 28, 2023 - Jul. 28, 2024 **VNU University of Science Basic-Level Grant**

- **Grant Number:** TN.23.04.
- **Project Title:** On the complexity of some reconfiguration problems on graphs under some distance constraints. (Độ phức tạp tính toán của một số bài toán tái cấu hình trong đồ thị với một số điều kiện về khoảng cách.)
- **Role:** Principal Investigator.

Aug. 30, 2019 - Mar. 31, 2021 **JSPS KAKENHI Grant-in-Aid for Research Activity start-up**

- **Grant Number:** 19K24349.
- **Project Title:** A study on reconfiguration problems under Token Sliding and their applications.
- **Role:** Principal Investigator.

## Awards

Jun. 22, 2018 JAIST Outstanding Performance Award for doctoral students.

## Research Visits

| | |
|---|---|
| Jan. 01, 2024 - Dec. 31, 2024 | Vietnam Institute for Advanced Study in Mathematics (Hanoi, Vietnam). Supported by VIASM's one-year postdoctoral fellowship. |
| Apr. 07, 2021 - Apr. 08, 2021 | Group of Computer Algorithms (Minato Lab), Graduate School of Informatics, Kyoto University (Kyoto, Japan).<br>Host: Shin-ichi MINATO. |
| Dec. 23, 2019 - Dec. 25, 2019 | Faculty of Advanced Science and Technology, Kumamoto University (Kumamoto, Japan).<br>Host: Yota OTACHI. |
| Apr. 01, 2016 - Jul. 08, 2016 | Algorithm Theory Lab, Graduate School of Information Sciences, Tohoku University (Sendai, Japan).<br>Host: Xiao ZHOU and Takehiro ITO. |

## Research Activities

Research Mentor
- Vietnam Polymath REU (2023-2024)

(Sub-)Reviewer

**Journal**
- Graphs and Combinatorics (2023)
- Electronic Journal of Combinatorics (2023)
- Journal of Information Processing (2020)
- Theoretical Computer Science (2018, 2019, 2021)
- Discrete Applied Mathematics (2018)
- IEICE TRANSACTIONS on Fundamentals of Electronics, Communications and Computer Sciences (2017, 2019)

**Conference**
- ICALP 2024
- ISAAC 2023
- ISAAC 2022
- WALCOM 2021, ICALP 2021, MFCS 2021
- WG 2020, COCOON 2020, ISAAC 2020
- MFCS 2019
- COCOON 2018

## Students

| | |
|---|---|
| Nguyễn Ánh Hồng | VNU University of Science, Hanoi (Oct. 2023 - Jun. 2024, Undergraduate Thesis) |
| Phan Minh Vũ | VNU University of Science, Hanoi (Oct. 2023 - Jun. 2024, Undergraduate Thesis) |
| Đào Xuân Thắng | VNU University of Science, Hanoi (Oct. 2023 - Jun. 2024, Undergraduate Thesis) |
| Phan Hữu An | Nanyang Technological University Singapore (Oct. 2023 - Oct. 2024, Vietnam Polymath REU) |
| Lâm Nhật Quân | HCMC University of Science, Vietnam (Oct. 2023 - Oct. 2024, Vietnam Polymath REU) |

## Teaching Experiences

A list of courses I have participated in as a **Lecturer** or **Teaching Assistant**. The courses in English (Vietnamese) are described in English (Vietnamese).

| | |
|---|---|
| Jan. 22, 2024 - Jun. 20, 2024 | **Lecturer** – VNU-HUS MAT2315: Phương pháp nghiên cứu khoa học (tiểu luận) |

| | |
|---|---|
| Jan. 22, 2024 - Jun. 20, 2024 | **Lecturer** – VNU-HUS MAT3500 2: Toán rời rạc |
| Jan. 22, 2024 - Jun. 20, 2024 | **Lecturer** – VNU-HUS MAT3500 1: Toán rời rạc |
| Feb. 6, 2023 - Jun. 02, 2023 | **Lecturer** – VNU-HUS MAT2315: Phương pháp nghiên cứu khoa học (tiểu luận) |
| Feb. 6, 2023 - Jun. 02, 2023 | **Lecturer** – VNU-HUS MAT3500 3: Toán rời rạc |
| Feb. 6, 2023 - Jun. 02, 2023 | **Lecturer** – VNU-HUS MAT3500 2: Toán rời rạc |
| Sep. 10, 2018 - Dec. 13, 2018 | **Lecturer** – VNU-HUS MAT3302: Toán rời rạc (bài tập) |
| Sep. 06, 2018 - Dec. 13, 2018 | **Lecturer** – VNU-HUS MAT3302 2TNT: Toán rời rạc (bài tập) |
| Oct. 11, 2017 - Nov. 30, 2017 | **Teaching Assistant** – JAIST I216: Computational Complexity and Discrete Mathematics. |
| Apr. 12, 2017 - Jun. 02, 2017 | **Teaching Assistant** – JAIST I216: Computational Complexity and Discrete Mathematics. |
| Oct. 12, 2016 - Dec. 01, 2016 | **Teaching Assistant** – JAIST I216: Computational Complexity and Discrete Mathematics. |
| Apr. 08, 2015 - Jun. 05, 2015 | **Teaching Assistant** – JAIST I216: Computational Complexity and Discrete Mathematics. |

## Co-authors (in alphabetical order)

David Avis, Erik D. Demaine, Martin L. Demaine, Eli Fox-Epstein, Takehiro Ito, Amanj Khorramian, Hirotaka Ono, Yota Otachi, Akira Suzuki, Ryuhei Uehara, Tsuyoshi Yagita, Takeshi Yamada.

## Publications

In all of my publications, I use the name **Duc A. Hoang**. A list of my publications can also be found at DBLP and Google Scholar. Some of them are available as e-prints at arXiv. All of my publications (including unpublished manuscripts) are listed at `https://hoanganhduc.github.io/publications/`.

### Journal


[6] David Avis and **Duc A. Hoang**. "A Note on Acyclic Token Sliding Reconfiguration Graphs of Independent Sets". In: *Ars Combinatoria* 159 (2024), pp. 133–154. DOI: `10.61091/ars159-12`.

[5] **Duc A. Hoang**. "On the complexity of distance-$d$ independent set reconfiguration". In: *Theoretical Computer Science* 1009 (2024). (article 114682). DOI: `10.1016/j.tcs.2024.114682`.

[4] David Avis and **Duc A. Hoang**. "On Reconfiguration Graphs of Independent Sets under Token Sliding". In: *Graphs and Combinatorics* 39.3 (2023). (article 59). DOI: `10.1007/s00373-023-02644-w`.

[3] **Duc A. Hoang**. "TS-Reconfiguration of $k$-Path Vertex Covers in Caterpillars for $k \geq 4$". In: *Theory and Applications of Graphs* 10.1 (2023). (article 8). DOI: `10.20429/tag.2023.10108`.

[2] **Duc A. Hoang**, Akira Suzuki, and Tsuyoshi Yagita. "Reconfiguring $k$-path vertex covers". In: *IEICE Transactions on Information and Systems* E105.D.7 (2022), pp. 1258–1272. DOI: `10.1587/transinf.2021EDP7177`.

[1] Erik D. Demaine, Martin L. Demaine, Eli Fox-Epstein, **Duc A. Hoang**, Takehiro Ito, Hirotaka Ono, Yota Otachi, Ryuhei Uehara, and Takeshi Yamada. "Linear-time algorithm for sliding tokens on trees". In: *Theoretical Computer Science* 600 (2015), pp. 132–142. DOI: `10.1016/j.tcs.2015.07.037`.


### Refereed International Conference


[7] **Duc A. Hoang**. "On the complexity of distance-$d$ independent set reconfiguration". In: *Proceedings of WALCOM 2023*. Ed. by Bertrand M. T. Lin, Chun-Cheng Lin, and Giuseppe Liotta. Vol. 13973. LNCS. Springer, 2023, pp. 254–266. DOI: `10.1007/978-3-031-27051-2_22`.

[6] **Duc A. Hoang**, Akira Suzuki, and Tsuyoshi Yagita. "Reconfiguring $k$-path vertex covers". In: *Proceedings of WALCOM 2020*. Ed. by M. Sohel Rahman, Kunihiko Sadakane, and Wing-Kin Sung. Vol. 12049. LNCS. Springer, 2020, pp. 133–145. DOI: `10.1007/978-3-030-39881-1_12`.

[5] **Duc A. Hoang**, Amanj Khorramian, and Ryuhei Uehara. "Shortest reconfiguration sequence for sliding tokens on spiders". In: *Proceedings of CIAC 2019*. Ed. by Pinar Heggernes. Vol. 11485. LNCS. Springer, 2019, pp. 262–273. DOI: 10.1007/978-3-030-17402-6_22.

[4] **Duc A. Hoang**, Eli Fox-Epstein, and Ryuhei Uehara. "Sliding tokens on block graphs". In: *Proceedings of WALCOM 2017*. Ed. by Sheung-Hung Poon, Md. Saidur Rahman, and Hsu-Chun Yen. Vol. 10167. LNCS. Springer, 2017, pp. 460–471. DOI: 10.1007/978-3-319-53925-6_36.

[3] **Duc A. Hoang** and Ryuhei Uehara. "Sliding tokens on a cactus". In: *Proceedings of ISAAC 2016*. Ed. by Seok-Hee Hong. Vol. 64. LIPIcs. Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, 2016, 37:1–37:26. DOI: 10.4230/LIPIcs.ISAAC.2016.37.

[2] Eli Fox-Epstein, **Duc A. Hoang**, Yota Otachi, and Ryuhei Uehara. "Sliding token on bipartite permutation graphs". In: *Proceedings of ISAAC 2015*. Ed. by Khaled Elbassioni and Kazuhisa Makino. Vol. 9472. LNCS. Springer, 2015, pp. 237–247. DOI: 10.1007/978-3-662-48971-0_21.

[1] Erik D. Demaine, Martin L. Demaine, Eli Fox-Epstein, **Duc A. Hoang**, Takehiro Ito, Hirotaka Ono, Yota Otachi, Ryuhei Uehara, and Takeshi Yamada. "Polynomial-time algorithm for sliding tokens on trees". In: *Proceedings of ISAAC 2014*. Ed. by Hee-Kap Ahn and Chan-Su Shin. Vol. 8889. LNCS. Springer, 2014, pp. 389–400. DOI: 10.1007/978-3-319-13075-0_31.

## Thesis/Dissertation

[2] **Duc A. Hoang**. "Independent set reconfiguration and related problems for some restricted graphs". PhD thesis. Japan Advanced Institute of Science and Technology, June 2018. URL: http://hdl.handle.net/10119/15431.

[1] **Duc A. Hoang**. "The independent set reconfiguration problem on some restricted graphs". MA thesis. Japan Advanced Institute of Science and Technology, Mar. 2015. URL: http://hdl.handle.net/10119/12643.